Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Zheng, Chiang, and Sheng et al. (2023)

Hins Fang Charles Costanzo

Motivation: Evaluating LLMs is hard!

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Tried before, but never studied systematically.



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That is, how well do LLM judges match **human preferences**?

- (and how do we even **measure** that?)

Meet the models: camelids in the hot seat



LLaMA-13B (Meta, 2023)



Vicuna-13B (LMSYS, 2023)

- The paper evaluates other chatbots as well, including GPT-4 and Claude.

Photos: PA (llama) and Marshallhenrie (vicuña), Wikimedia Commons, CC BY-SA 4.0

Meet the judges: LLMs on the bench

GPT-4



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- All three are checked against **human** judges:
  - 58 experts (MT-Bench) and crowdsourced ratings (Chatbot Arena)
- The judges are also among the models evaluated, so they sometimes grade their own answers.

Example 1:

Question

If the FED buys bonds in the secondary market

(A) the money supply will increase.
(B) the money supply will decrease.
(C) the money supply will not be affected.
(D) the discount rate would be affected.

Output the answer only.

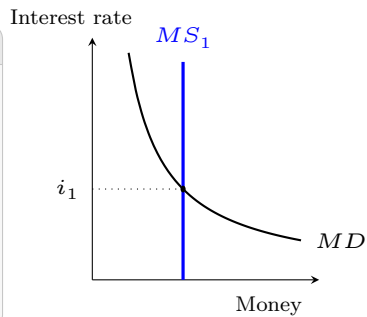
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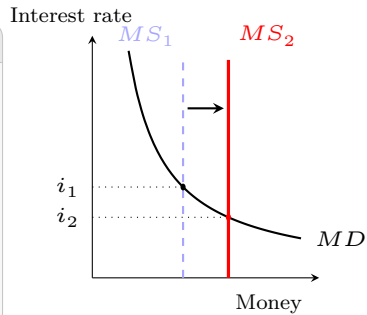
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Example 1

LLaMA-13B

```
The Federal Reserve buys bonds in the secondary
market to increase the money supply.
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Vicuna-13B

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Example 1 - Follow-up:

Follow-up Question

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How does it affect my daily life? Give 3 examples

LLaMA-13B

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:

1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

Vicuna-13B

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

You be the judge



menti.com code **1479 0094**

Which assistant did better in this conversation?

- LLaMA-13B
- Vicuna-13B
- Tie

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Then we'll see what GPT-4 said.

Example 1 - GPT-4 as judge

GPT-4 Judgment

LLaMA-13B provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.

On the other hand, Vicuna-13B provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

LLaMA-13B should improve by providing relevant, clear, and accurate examples to address the user's question. **Vicuna-13B's response is more helpful, relevant, and detailed.**

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Vicuna more helpful, relevant, and detailed

- LLaMA: repetitive, no real examples
- Vicuna: three clear, relevant examples

Scores under different benchmarks

Example 1 is an MMLU[1] question. Both answered correctly, but LLaMA ignored "answer only."

Model	MMLU (5-shot)	MT-Bench score (GPT-4)
LLaMA-13B	47.0	2.61
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MMLU was not designed to measure human preferences.

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Key topics

- **Ground truth:** collecting human preferences to evaluate LLM judges against
  - **MT-Bench**: 80 multi-turn questions, 3,000 expert votes
  - **Chatbot Arena**: crowdsourced anonymous battles, 30,000 conversations

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- **Limitations:** LLM biases and reasoning errors
  - As well as strategies to address some of them
- **Agreement:** how often LLM judges match human preferences

Two new benchmarks

- Recall Example 1:
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- Recall Example 1:
  - MMLU barely separated LLaMA and Vicuna
  - But humans strongly preferred Vicuna
- Most existing benchmarks measure **correctness**
  - e.g., MMLU: multiple choice, checked against an answer key
- Chatbots are used for **open-ended**, **multi-turn** conversations
  - No single right answer to check against

Two new benchmarks

New primary evaluation metric:
human ratings

MT-Bench · Chatbot Arena

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- Judged by **58 experts**, given nearly the same instructions as GPT-4

MT-Bench: what a question looks like

Writing, turn 1

Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.

Math, turn 1

Given that $f(x) = 4x^3 - 9x - 14$, find the value of $f(2)$.

Questions from Table 1, Zheng et al. (2023).

MT-Bench: what a question looks like

Writing, turn 1

Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.

Writing, turn 2

Rewrite your previous response. Start every sentence with the letter A.

Math, turn 1

Given that $f(x) = 4x^3 - 9x - 14$, find the value of $f(2)$.

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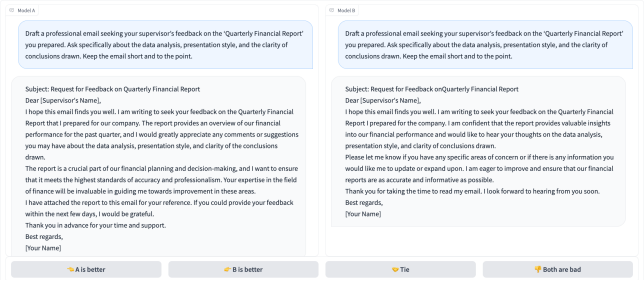
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Following a new instruction while building on its **own** previous answer.

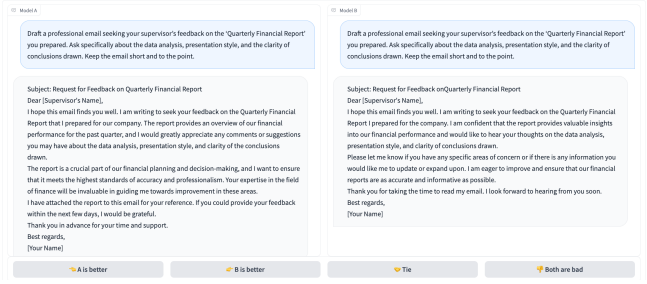
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Chatbot Arena: battles in the wild



Screenshot cropped from Figure 19 in Zheng et al. (2023).

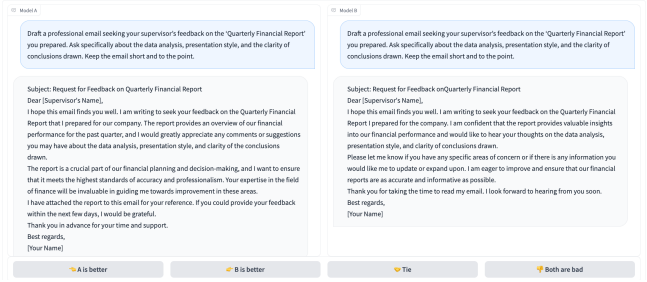
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- Users ask anything, see two **anonymous** answers, and vote. Names are revealed only afterwards.
- About 30,000 votes in the first month.

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Chatbot Arena: battles in the wild



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- About 30,000 votes in the first month.
- Your poll was a mini Arena, except you knew which model was which. **Why hide the names?**

Screenshot cropped from Figure 19 in Zheng et al. (2023).

Two kinds of ground truth

	MT-Bench	**Chatbot Arena**
Questions	Fixed, curated	Whatever users ask
Judges	58 experts	Crowd
Votes	About 3,000	About 30,000
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Now we have human verdicts. Can an LLM reproduce them, and where does it go wrong?

How an LLM judge is asked

- **Pairwise:** question + two answers → "A is better," "B is better," or tie
- **Single answer:** question + one answer → a score from 1 to 10
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Answers are labeled only "Assistant A" and "Assistant B."
The judge is **never told** which model wrote which answer.

Four limitations

1. **Position bias:** favors an answer because of where it appears
2. **Verbosity bias:** favors longer answers, even with nothing added
3. **Self-enhancement bias:** favors its own answers
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The default prompt already says to ignore position, length, and names.

Position bias

- Test: two near-identical answers to each of 80 questions, judged in both orders
- A consistent judge picks the same answer after the swap

Table 2, default prompt, Zheng et al. (2023).

Position bias

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Judge	Consistent after swap	Favored first answer
Claude-v1	23.8%	75.0%
GPT-3.5	46.2%	50.0%
GPT-4	65.0%	30.0%

- Most judges favor the **first** answer. Even GPT-4 is inconsistent a third of the time.

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Verbosity bias

- "Repetitive list" attack on 23 answers that contain a list
- Rephrase the list and paste it in front: twice as long, **no new information**
- Does the judge prefer the padded version?

Table 3, Zheng et al. (2023). Example in Figure 12.

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	Claude-v1	GPT-3.5	GPT-4
Fooled by the padding	91.3%	91.3%	8.7%

Table 3, Zheng et al. (2023). Example in Figure 12.

Self-enhancement bias

- Versus human win rates, GPT-4 rates itself about **10%** higher and Claude-v1 about **25%**
- GPT-3.5 does **not** favor itself

Figure 3(b), non-tied first-turn votes, Zheng et al. (2023).

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- And what would it even mean?
  - **Recognition:** knows the answer is its own, but is never told
  - **Style preference:** likes its own writing style

Figure 3(b), non-tied first-turn votes, Zheng et al. (2023).

How strong is the evidence?

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- A 10-point gap in win rate could be noise
- **Your turn:** how would you test recognition vs. style preference?

Limited reasoning: misled by the answers

Question

Benjamin bought 5 sci-fi novels at $20 each, 3 history books at $30 each, and 2 philosophy books at $45 each. What was the total cost?

Assistant A

The total cost of Benjamin's purchases was $115.

Assistant B

The total cost of his purchases is $295.

Question shortened from Figure 13, Zheng et al. (2023).

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GPT-4 judgment

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The correct total is $280, so **both** answers are wrong. Asked on its own, GPT-4 gets $280.

Question shortened from Figure 13, Zheng et al. (2023).

What the paper fixes

- **Position bias:** judge both orders; a win must hold in both, otherwise it is a tie

Table 4: 10 questions, each judged in both orders, Zheng et al. (2023).

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Verbosity and self-enhancement: no fix proposed.

Table 4: 10 questions, each judged in both orders, Zheng et al. (2023).

References

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